

NUTRITION FOR DEVELOPING SWIM ATHLETES

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Caloric Needs

- Calories = Energy
- Deficit - performance and development will suffer
- Dependent on the individual and type of training that day
- Big idea: need energy for exercise itself, to pay back oxygen debt, and to repair/rebuild



MACRONUTRIENTS

Why Do They Matter?

- Macronutrients affect many processes in our body, including (but not limited to):
 - Our immune system and health
 - Our hormone production
 - Our ability to digest and absorb nutrients
 - Our body composition
 - Our cell's structure and function
 - Our metabolic function



Carbohydrates

- Fastest-acting macronutrient source for energy transfer
- Glycogen stores are limited and need replenished
- Simple Carbohydrate – faster digesting, usually not as healthy
 - Appropriate in-between races to replenish glycogen levels quickly
 - Will go to replenishment and oxidation, rather than storage
 - **Examples: energy bar, instant oatmeal, fruit snacks, cereal, bagel, rice cake**
- Complex Carbohydrate – slower digesting, usually a healthier option
 - Day-to-day nutrition should revolve around these

Examples of Complex Carbohydrates

- Fruits
- Vegetables
- Sprouted & whole grains
 - Oats
 - Millet
 - Buckwheat
 - Barley
 - Brown rice
- Sweet potatoes
- Quinoa



Anti-Inflammatory Fats

- Provide energy for low-moderate intensity exercise
- Assist in vitamin absorption
- Lower inflammation, support immune function
- Help regulate hormones, facilitate nerve actions
- Support brain health & function
- Omega-3 & Omega-6 have to come from our diet
 - A varied diet will help balance these
 - Omega-3 can come from a fish oil supplement
- Wild-caught and pasture-raised animal products may provide improved fatty acid profiles

Examples of Healthy Fats

- Avocados
- Dairy
- Eggs
- Fatty fish (wild-caught)
- Poultry, beef, pork, lamb, wild game
- Olives and extra-virgin olive oil
- Brazil nuts, macadamia nuts, walnuts, pecans, almonds, cashews, pistachios
- Chia, hemp, pumpkin, sunflower, sesame and flax seeds
- Cacao
- Coconut



Protein

- Building blocks
 - The body is in a constant state of breaking down and building up new proteins
- Essential for muscle growth and repair
- Made up of amino acids
 - Need 9 essential amino acids that only come from protein in the form of food
 - Our bodies need protein and amino acids to produce important molecules that we can't function without
- We can't store protein in the same way we store carbs or fats
- Not all proteins are created equal

Examples of High Protein Foods

- Chicken, beef
- Wild caught fish
- Eggs
- Dairy such as cottage cheese or Greek yogurt
- Tofu, Tempeh
- Whey protein isolate



MICRONUTRIENTS

Am I at a Deficit?

- Could be getting all of your macronutrients, but be at a deficit on all of your micronutrients
- Not everyone needs the same micronutrients in the same amounts
- In general, athletes will need more micronutrients than the average person
- Includes vitamins, minerals, phytonutrients, myconutrients, zoonutrients
- Healthy metabolism, repair & restoration of tissues, digestion, energy transfer, nervous system function
- Quality multi-vitamin or targeted supplementation an option to fill in the gaps



PRE-PUBERTY

Nutrition for Proper Development

- Brain
 - Dietary fat
 - Choline
 - Folic acid
 - Magnesium
 - Zinc
 - B12
- Muscle
 - Protein
 - Vitamin B6
 - Vitamin B9
 - Calcium
 - Vitamin D

Nutrition for Proper Development Cont.

- Immune System
 - Dietary fats
 - Protein
 - Vitamin C
 - Zinc

- Bone
 - Vitamin D
 - Vitamin C
 - Vitamin K₁ + K₂
 - Calcium

Nutrition for Proper Development Cont.

- Cardiovascular System

- Dietary fats
- Iron

- Nervous System

- Vitamin B6
- Vitamin B12
- Zinc

- Hormones

- Dietary fats
- Protein
- Vitamin B6
- Iodine
- Zinc

Common Deficiencies in Children

- Iron – beans, meat, whole grains, greens
- Zinc – beans, meat, whole grains, crab
- Magnesium – pumpkin seeds, black beans, brown rice, chia seeds, avocados, leafy greens
- Calcium – beans, greens, nuts, seeds
- Folic Acid – whole grains, beans, fruits, vegetables
- Vitamin A – orange/ yellow veggies, leafy greens, tomatoes, cantaloupe, milk, eggs
- Vitamin C – strawberries, bell peppers, oranges, cantaloupe, kiwi, pineapple
- Vitamin D – fish, eggs, dairy, mushrooms, fortified foods
- Vitamin B6 – whole grains, fruits, vegetables, beans, meats
- Vitamin B12 – animal foods
- Iodine – iodized salt, dairy, fish, sea vegetables



Deficiency Implications

- Vitamin D
 - Rickets, deformed bones, slowed growth, soft teeth
- Iodine
 - Impaired growth and neurological development
- Iron
 - Behavioral abnormalities, low immunity
- Zinc
 - Delayed growth and sexual maturation, skeletal abnormalities

Strategies for Success

- Take the lead
- Have a variety of unprocessed and less-processed whole foods available
- Keep ultra-processed options to a minimum
- Give them choices
- Get them involved in the preparation/menu planning
- Make meal time family time
- Slow down!



POST-PUBERTY

The Basics + Some

- Pre-puberty “big picture” nutrition practices
 - Whole, minimally processed most of the time
- Increased caloric need
- Increased specific need for protein, iron, calcium, zinc, folate



The Basics + Some Cont.

- Protein recommendation for athletes
 - 0.7-1 g of protein per pound of body weight
- Iron
 - Dark leafy greens, beans, legumes, olives, raisins, whole grains, peppers, potatoes, seeds, prune juice, fish, shellfish, organ meats, red meats, dark poultry
- Calcium
 - Dairy, dark green vegetables, beans, nuts, seeds, rhubarb, fish
- Zinc
 - Beans, legumes, nuts, seeds, whole grains, seafood, beef, lamb, pork, poultry, eggs, wild game, mushrooms
- Folate
 - Chickpeas, pinto beans, liver, lentils, spinach, asparagus, beets, avocado, broccoli

WATER

Water Intake

- Athletic needs are greater
- Think, focus, and perform better
- Baseline need
 - Minimum $\frac{1}{2}$ bodyweight in ounces to start (will likely need more)
- Sweat around 0.5-2.0 L per hour of activity (16 – 67 oz)
- **General principle:** aim for half your bodyweight in ounces + an additional 16-32 oz per hour of activity



COUNTING THE COST

Inconsistency

- Development issues
- Failure to recover or refuel properly
- Brain “fog”
- Sickness due to weakened immune system
- Low energy levels/sluggish
- Depression & anxiety
- **Inconsistency = Poor performance aka SLOW swimming**

REAPING THE BENEFIT

Consistency

- New and healthy tissue created
- Proper development
- Fuel stores rejuvenated for next training session/competition
- Proper circulation of oxygen
- Mental clarity
- **Consistency = Improved performance aka FAST swimming**



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